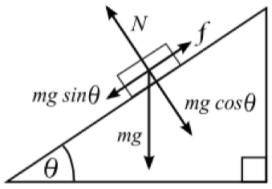
# 1 Introduction

Sometimes a given basis is not the most convenient for a particular problem, and it would be better to change to another basis. Consider, for example, the motion of an object down an inclined plane angled $\theta = 30^\circ$ above the ground.




Since the basis vectors of the standard basis of $\mathbb{R}^2$ are parallel and perpendicular to the ground, there is motion in both of these directions, so Newton's second law gives rise to a coupled system of differential equations. It is much more convenient to choose a rotated coordinate system, which corresponds to choosing the basis

$$v_{\parallel} = (\cos(30^\circ), \sin(30^\circ)) = (\sqrt{3}/2, 1/2),$$
$$v_{\perp} = (\sin(30^\circ), \cos(30^\circ)) = (-1/2, \sqrt{3}/2),$$

where $v_{\parallel}$ points down the plane and $v_{\perp}$ is perpendicular to the plane. It is much easier to solve the equations of motion in this basis since the motion is only along the $v_{\parallel}$ direction.

# 2 Change of Basis - Coordinate Vectors

Let $V$ be a finite-dimensional vector space and let $B = (v_1, \ldots, v_n)$ be an ordered basis of $V$. Then for any $v \in V$, there exist unique scalars $\alpha_1, \ldots, \alpha_n$ such that

$$v = \alpha_1 v_1 + \cdots + \alpha_n v_n. \tag{2.1}$$

Let $B' = (w_1, \ldots, w_n)$ be another ordered basis of $V$. There is another set of unique scalars $\beta_1, \ldots, \beta_n$ such that

$$v = \beta_1 v_1 + \cdots + \beta_n v_n. \tag{2.2}$$

We would like to know how the coordinate vector $[v]_{B'} = (\beta_1, \ldots, \beta_n)$ is obtained from $[v]_B = (\alpha_1, \ldots, \alpha_n)$ by changing basis from $B$ to $B'$.

Since can write each vector $v \in V$ as a linear combination of the vectors in $B'$. In particular, this is true for each vector in $B$. That is, for each $j = 1, \ldots, n$, there exist unique scalars $C_{ij}$ such that

$$v_j = C_{1j} w_1 + \cdots + C_{nj} w_n. \tag{2.3}$$

Substituting (2.3) into (2.1) for each $v_j$ gives

$$\begin{aligned} v &= \sum_{j=1}^{n} \alpha_j \sum_{i=1}^{n} C_{ij} w_i \\ &= \sum_{i=1}^{n} \left( \sum_{j=1}^{n} C_{ij} \alpha_j \right) w_i. \end{aligned}$$

Comparing this with (2.2), we see that

$$\beta_i = \sum_{j=1}^{n} C_{ij} \alpha_j.$$

That is to obtain the new coordinate vector $[v]_{B'}$, we simply multiply the old coordinate vector $[v]_B$ by the matrix $C$, whose $j$th column is $[v_j]_{B'}$ (the coordinate vector of the $j$th old basis vector with respect to the new basis $B'$). The matrix $C$ is called the *transition matrix* or *change of basis matrix* from $B$ to $B'$.

By interchanging the roles of $B$ and $B'$, it follows immediately that $C$ is invertible, with $C^{-1}$ being the transition matrix from $B'$ to $B$.

**Example 2.4.** Let $B = (v_1, v_2) = ((1,0),(0,1))$ be the standard basis of $\mathbb{R}^2$ and $B' = (w_1, w_2) = (\sqrt{3}/2, 1/2), (-1/2, \sqrt{3}/2))$ be the rotated basis from the example discussed in the first section. To find the matrix $C$, we expand each basis vector $v_j$ of $B$ in the basis $B'$.

$$\begin{aligned} v_1 &= C_{11} w_1 + C_{21} w_2 \\ (1,0) &= C_{11}(\sqrt{3}/2, 1/2) + C_{21}(-1/2, \sqrt{3}/2) \\ (1,0) &= \left( \frac{\sqrt{3}}{2} C_{11} - \frac{1}{2} C_{21}, \frac{1}{2} C_{11} + \frac{\sqrt{3}}{2} C_{21} \right) \end{aligned}$$

which implies the system of linear equations

$$\begin{aligned} \frac{\sqrt{3}}{2} C_{11} - \frac{1}{2} C_{21} &= 1 \\ \frac{1}{2} C_{11} + \frac{\sqrt{3}}{2} C_{21} &= 0 \end{aligned}$$

The second column is given by solving

$$\begin{aligned} v_2 &= C_{21} w_1 + C_{22} w_2 \\ (0,1) &= C_{21}(\sqrt{3}/2, 1/2) + C_{22}(-1/2, \sqrt{3}/2) \\ (0,1) &= \left( \frac{\sqrt{3}}{2} C_{21} - \frac{1}{2} C_{22}, \frac{1}{2} C_{21} + \frac{\sqrt{3}}{2} C_{22} \right) \end{aligned}$$

which implies the system of linear equations

$$\begin{aligned} \frac{\sqrt{3}}{2} C_{21} - \frac{1}{2} C_{22} &= 0 \\ \frac{1}{2} C_{21} + \frac{\sqrt{3}}{2} C_{22} &= 1. \end{aligned}$$

Notice that these two systems have exactly the same coefficient matrix. We can therefore solve both systems simultaneously by row-reduction by forming an augmented matrix with two additional columns $\begin{bmatrix}1\\0\end{bmatrix}$ and $\begin{bmatrix}0\\1\end{bmatrix}$:

$$\left[\begin{array}{cc|cc} \frac{\sqrt{3}}{2} & -\frac{1}{2} & 1 & 0 \\ \frac{1}{2} & \frac{\sqrt{3}}{2} & 0 & 1 \end{array}\right] \tag{2.5}$$

Row-reducing gives

$$\left[\begin{array}{cc|cc} 1 & 0 & \frac{\sqrt{3}}{2} & \frac{1}{2} \\ 0 & 1 & -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{array}\right],$$

and therefore

$$C = \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}.$$

*Remark* 2.6. (a) More generally, to find the transition matrix $C$ from basis $B$ to $B'$, we row-reduce the augmented matrix $[[w_i]|[v_i]]$, whose first $n$ columns are the coordinate vectors $[w_i]$ of the vectors in $B'$ and whose last $n$ columns are the coordinate vectors $[v_i]$ of the vectors in $B$.

(b) Row-reducing the augmented matrix in (2.5), as we learned last time, is precisely the technique for computing the inverse, so

$$C^{-1} = \begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$$

is the transition matrix from $B'$ to $B$. We can therefore view any nonsingular $n \times n$ matrix $A$ as the transition matrix from the ordered basis $B'$ of $F^n$ consisting of the columns of $C$ to the standard ordered basis of $F^n$.

**Example 2.7.** Let $B, B'$ be ordered bases for $F^3$ where $v_1 = (2,1,1),\ v_2 = (2,-1,1),\ v_3 = (1,2,1)$ and $w_1 = (3,1,-5),\ w_2 = (1,1,-3),\ w_3 = (-1,0,2)$. To compute the transition matrix $C$ from $B$ to $B'$, we row-reduce the augmented matrix

$$[[w_i]|[v_i]] = \left[\begin{array}{ccc|ccc} 3 & 1 & -1 & 2 & 2 & 1 \\ 1 & 1 & 0 & 1 & -1 & 2 \\ -5 & -3 & 2 & 1 & 1 & 1 \end{array}\right] \mapsto \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 3 & 2 & \frac{5}{2} \\ 0 & 1 & 0 & -2 & -3 & -\frac{1}{2} \\ 0 & 0 & 1 & 5 & 1 & 6 \end{array}\right]$$

and therefore $C = \begin{bmatrix} 3 & 2 & \frac{5}{2} \\ -2 & -3 & -\frac{1}{2} \\ 5 & 1 & 6 \end{bmatrix}$.

# 3 Change of Basis - Matrix of a Linear Operator

Let $T \in \mathscr{L}(V)$ and $B, B'$ two ordered bases of a finite-dimensional vector space $V$. Let $w = Tv$. In the coordinates defined by $B$, this becomes

$$[w]_B = [T]_B[v]_B. \tag{3.1}$$

We found in the previous section that $[w]_{B'} = C[w]_B$ and $[v]_{B'} = C[v]_B$, so if we instead use the coordinates defined by $B'$ we have

$$\begin{aligned}[w]_{B'} &= [T]_{B'}[v]_{B'} \\ C[w]_B &= [T]_{B'}C[v]_B.\end{aligned}$$

Multiplying both sides by $C^{-1}$ then gives

$$[w]_B = C^{-1}[T]_{B'}C[v]_B. \tag{3.2}$$

Comparing (3.2) and (3.1), we see that $[T]_B$ and $[T]_{B'}$ are related by

$$C^{-1}[T]_{B'}C = [T]_B$$

or, multiplying on the left by $C$ and the right by $C^{-1}$,

$$[T]_{B'} = C[T]_BC^{-1}. \tag{3.3}$$

**Example 3.4.** Consider the linear map $T : \mathbb{R}^2 \to \mathbb{R}^2$ defined by $T(x,y) = (2x-y, -3x+2y)$. With respect to the standard ordered basis $B$ of $\mathbb{R}^2$, the matrix of $T$ is given by

$$[T]_B = \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix}.$$

If we change basis to the ordered basis $B' = \{(\sqrt{3}/2, 1/2), (-1/2, \sqrt{3}/2)\}$, we found in Example 2.4 that

$$C = \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix} \quad \text{and} \quad C^{-1} = \begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix},$$

and therefore the matrix of $T$ with respect to the new ordered basis $B'$ is

$$\begin{aligned}[T]_{B'} &= C[T]_BC^{-1} \\ &= \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix} \begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix} \\ &= \begin{bmatrix} 2-\sqrt{3} & 0 \\ -2 & 2+\sqrt{3} \end{bmatrix}.\end{aligned}$$

# 4 Exercises

1. Consider the ordered bases $B = (v_1, v_2)$ and $B' = (w_1, w_2)$ of $F^2$ where

$$v_1 = (2,2), \quad v_2 = (4,-1),$$
$$w_1 = (1,3), \quad w_2 = (-1,-1).$$

   (a) Find the transition matrix from $B$ to $B'$.

   (b) Find the transition matrix from $B'$ to $B$.

   (c) Given $u = (3,-5)$ compute the coordinate vectors $[u]_B$ and $[u]_{B'}$.

   (d) Given the linear operator $T(x,y) = (x+2y, -4x-2y)$, calculate $[T]_B$ and $[T]_{B'}$.

2. Let $V = \text{span}_{\mathbb{R}}\{f_1, f_2\}$, where $f_1(x) = \sin(x)$ and $f_2(x) = \cos(x)$. Let $B = (f_1, f_2)$ and $B' = (g_1, g_2)$, where $g_1(x) = 2\sin(x) + \cos(x)$ and $g_2 = 3\cos(x)$.

   (a) Find the transition matrix from $B$ to $B'$.

   (b) Find the transition matrix from $B'$ to $B$.

   (c) Given $h(x) = 2\sin(x) - 5\cos(x)$ compute the coordinate vectors $[h]_B$ and $[h]_{B'}$.

   (d) Given the linear operator $T(c_1\sin(x) + c_2\cos(x)) = (c_2 - c_1)\sin(x)$, calculate $[T]_B$ and $[T]_{B'}$.

3. Let $V = \mathscr{P}_1(F)$, $B = (p_1, p_2)$, $B' = (q_1, q_2)$, where

$$p_1 = 6 + 3x, \quad p_2 = 10 + 2x,$$
$$q_1 = 2, \quad q_2 = 3 + 2x.$$

   (a) Find the transition matrix from $B$ to $B'$.

   (b) Find the transition matrix from $B'$ to $B$.

   (c) Given $p = -4 + x$ compute the coordinate vectors $[p]_B$ and $[p]_{B'}$.

   (d) Given the linear operator $T(a + bx) = 3a + 2bx$, calculate $[T]_B$ and $[T]_{B'}$.